

Want to discover unknown classes in your data?



Motivation for OmniGCD

1. In the human brain, the prefrontal cortex integrates highly preprocessed signals from other major forebrain areas^[1]
2. Recent studies show that category formation and representation in humans is independent of any specific sensory inputs^[2]
3. GCD works currently only focus on category discovery within a single isolated domain (e.g. vision or text)

The problem

Existing GCD works focus on GCD within **one** isolated modality and require independent **fine-tuning** on each dataset.

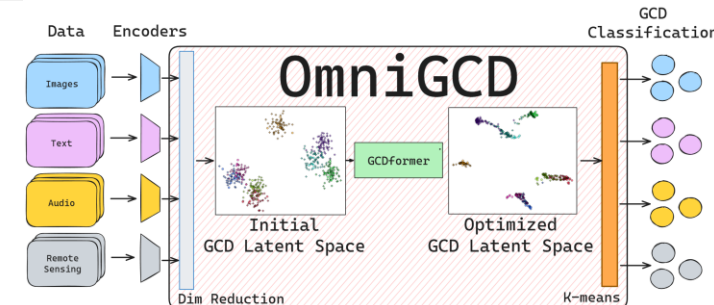
*Are these methods truly demonstrating **Generalized** category discovery?*

What is the GCD task?

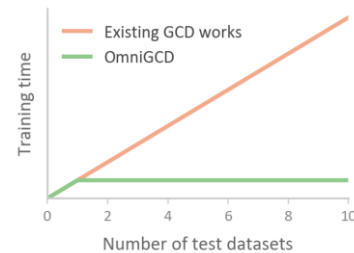
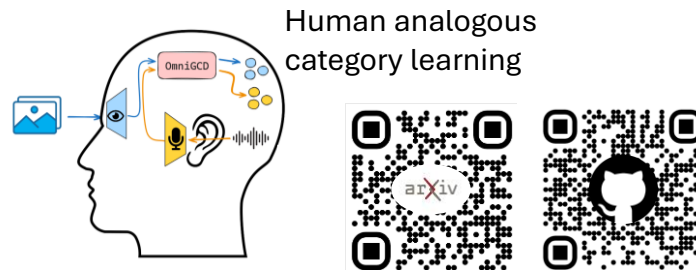
In **Generalized Category Discovery (GCD)** the task is to **assign labels to an unknown set of images D_U given some known labelled set of images D_L** ; where $D_L \subset D_U$.

Standard GCD^[3]: Allow access to D_U and D_L for training

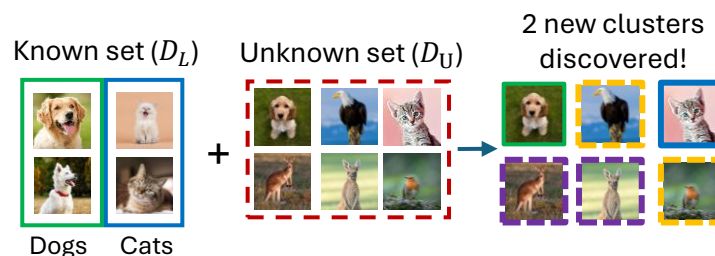
Zero-shot GCD: No access to D_U and D_L for training, D_L accessed at test time to predict labels in D_U



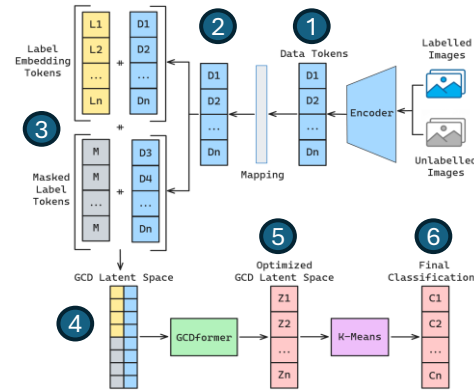
OmniGCD pairs with modality specific encoders to perform the GCD task using **one model on any modality**



Train OmniGCD **once** on the synthetic GCD task and use/re-use with any modality specific encoder.



How it works



1. Compute feature vectors of labelled and unlabelled data (**data tokens**)
2. Map **data tokens**, via dimension reduction, to a common dimensionality
3. Concatenate **data tokens** with embedded label information (**label tokens**)
 - Label embedding token for known
 - Masked label token for unknown
4. Concatenate all tokens to form a **GCD latent space**
5. **GCDformer** optimises the **GCD latent space** for clustering
6. K-means produces **final classifications**

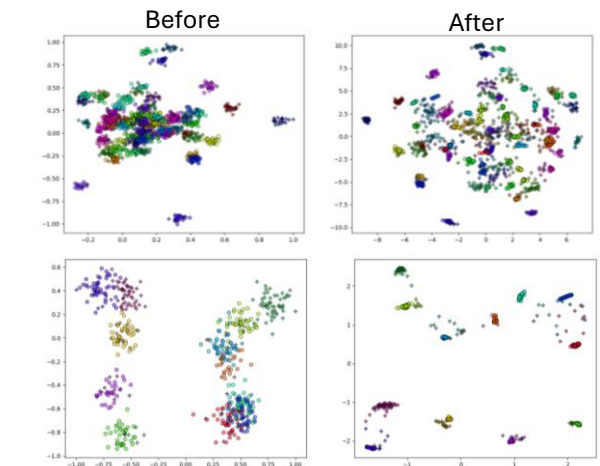
Zero-shot GCD Results

OmniGCD performs zero-shot GCD across 16 datasets spanning 4 modalities.



Optimized GCD Latent Spaces

GCDformer optimizes the GCD latent space for clustering (●: labelled points, +: unlabelled)



Acknowledgements

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References

- [1] Earl K. Miller and Jonathan D. Cohen. An integrative theory of prefrontal cortex function. Annu. Rev. Neurosci., 2001
- [2] Julien Dirani and Liina Pykkänen. Meg evidence that modality-independent conceptual representations contain semantic and visual features. J. Neurosci., 2024
- [3] Sagar Vaze, Kai Han, Andrea Vedaldi, and Andrew Zisserman. Generalized category discovery. CVPR, 2022.